



VTPEH 6105 – BIOSTATISTICS FOR HEALTH SCIENCES

Syllabus – Spring 2026

Course Details

VTPEH 6105	Biostatistics for Health Sciences, 3 credits
Term	Spring Semester
Class meets	Friday, 11:15-01:45 PM (one 2h30 session)
Workload	14 weeks with: - In class lecture and practice sessions: 2h30 per week - Homework: approximately 5 hours per week
Location	LH3 (S2-223) or LH4 (S1-210) as indicated on Canvas

Faculty Contact Information

Amandine Gamble
Department of Public and Ecosystem Health
amandine.gamble@cornell.edu

Course Assistants

Jordan Deskins – jad537@cornell.edu – Mac user
Anika Jessup – awj32@cornell.edu – Mac user
Mainul Islam – mi298@cornell.edu – Mac user
Princal Patel – pp528@cornell.edu – PC user
Madison Wolfinger – mhw88@cornell.edu – Mac user
Daniel Addo (VTPEH 6108) – da456@cornell.edu – PC user
Harsh Sonar (VTPEH 6108) – hs2262@cornell.edu – PC user
Therese Martin Cheteu Wabo (VTPEH 6108) – tmw95@cornell.edu
Dan Lin (VTPEH 6108) – dl862@cornell.edu

Guest Lecturers

Corinna Noel – can64@cornell.edu
Amelia Nelson – aln65@cornell.edu
Katherine Koebel – kjk239@cornell.edu

Office Hours

Amandine Gamble: by appointment via [Calendly](#)

Course Assistants:

- **Tuesday, 09:00-10:00 AM**, in **S1-122**, unless indicated otherwise on Canvas
With Jordan, Madison, Daniel and Harsh
- **Wednesday, 05:00-06:00 PM**, in **S1-122**, unless indicated otherwise on Canvas
With Mainul and Princal
- If you cannot make it at these times, please contact the course assistant of your choice by email

Course Description

This course provides a comprehensive introduction to **statistical inference**, equipping students with the skills to critically **understand and apply** statistical methods to **support evidence-based decision-making in Public Health**. Through interactive lectures and in-class practice, students will explore key concepts such as probability-based reasoning, hypothesis testing, and statistical fallacies. The course covers essential inferential techniques, including parametric and non-parametric statistical tests, regression models and multivariate analysis. This course provides an **introduction to R programming**, offering students an opportunity to build foundational computational skills. Assessment includes applied exercises to reinforce understanding of the assumptions and limitations of these methods, along with peer reviews of scientific publications, emphasizing the interpretation of statistical analyses. By the end of the course, students will have developed strong **statistical literacy skills essential for evaluating research and making informed decisions in public health practice**.

Additional support and opportunities for practice will be provided as part of VTPEH 6108 or VTPEH 6270.

Prerequisites

VTPEH 6104 – Epidemiology in Practice
VTPEH 6108 – Public Health Data Analysis Lab

Teaching Format

There will be 14 in-person sessions. Each session will consist of a **lecture** (~40 min) led by the instructor on a specific topic, an **applied workshop** (~40 min) and a **group working session** (~40 min) to prepare for the homework. All materials and information related to the course will be available on Canvas. All assignments must be submitted on Canvas.

Course Objectives

This course will address two core competencies of the MPH related to data analysis. By the end of the course, students will be able to:

- Run basic analyses to *analyze quantitative and qualitative data using biostatistics, informatics, computer-based programming, and software, as appropriate* (**CEPH Competency 3.3**).
- *Interpret data analysis to guide public health research, policy, or practice* (**CEPH Competency 3.4**).

These competencies will be developed through in-class workshops. They will be applied and assessed through content checks, data analysis exercises, and peer-reviews of scientific publications, as follow:

Learning Outcome	CEPH Competency	Method of Assessment
Analyze quantitative and qualitative data using biostatistics, informatics, computer-based programming, and software, as appropriate.	3.3	<i>Formative</i> - Content check quiz (in class) - Data analysis exercise (homework) - Guided peer-review (homework)
Interpret data analysis to guide public health research, policy, or practice.	3.4	<i>Summative</i> - Peer-review (final exam)

Assignments and Process of Assessment

Your learning and academic performance will be assessed through diverse formative and summative assignments. More details will be provided on each assignment on Canvas. Brief overviews are below.

Attendance and Participation in Class Activities *5% of final grade*

Participation in class activities is important as practice includes collective review and debugging sessions. Attend and participate in each class session. If you need to miss class, please let us know before the class session. Absences that are not excused will earn a zero for participation for that class.

Weekly Knowledge Checks *25% of final grade*

These short formative quizzes will be released prior to each class and will serve two purposes: (1) allow you to self-assess your mastery of recent course material, and (2) verify your readiness with prerequisite concepts needed for the next class session. Quizzes will be automatically graded with unlimited retake opportunities. Only your highest score will count toward your final grade.

Weekly Homework: Practice and Peer-Review *45% of final grade*

These formative assignments will focus on applying skills covered in class. These assignments will consist of a data analysis exercise (analysis of a dataset) and/or a peer-review (critical evaluation of a published analysis). These formative assessments will be graded on timeliness, understanding of the relevance of the analysis, calculation, understanding of the assumptions, and interpretation of the outputs.

Weekly Homework: Guiding Quizzes *5% of final grade*

These supporting quizzes will feature guiding questions to help approach and complete the weekly homework assignments. Released alongside each homework, participation requirements will be based on current homework performance: students above the grade threshold will be excused (but may participate for additional support or points), while those below the threshold will be required to complete the quiz to reinforce key concepts. Quizzes will be automatically graded with unlimited retake opportunities. Only the highest score will count toward the final grade.

Final Exam: Peer-Review 20% of final grade

This summative assignment will focus on applying skills covered over the course, with a focus on critical consideration of published studies, and interpretation of the outputs. This exam will consist of a peer-review (critical evaluation of a published analysis). This summative assessment will be graded on timeliness, understanding of the relevance of the analysis, understanding of the assumptions, and interpretation of the outputs.

Revisions

There will be no opportunity for revision.

Extensions and Late Work

Timeliness is included in each assignment grade. Assignments that are late without an extension will lose the timeliness point(s), as well as 5% of possible points per additional day. Extensions may be given (at instructor's discretion) if requested before the due date.

Final Course Grade

The final course grade is calculated based on all the assignments listed above. Letter grades are determined using the following scale:

Letter	Grade (%)	Point
A+	96-100	4.3
A	93-95	4
A–	90-92	3.7
B+	86-89	3.3
B	83-85	3
B–	80-82	2.7
C+	76-79	2.3
C	73-75	2
C–	70-72	1.7
D+	66-69	1.3
D	63-65	1
D–	60-62	0.7
F	<60	0

MPH Competence Assessment

Per the MPH Student Handbook, to be in good academic standing, a student in the MPH Program must:

1. Earn at least a grade of C or 73% in each course;
2. Be assessed as 'competent' in each relevant area assessed in each class each semester;
3. And maintain a minimum semester grade point average (GPA) of 2.5.

It is your responsibility to work hard and achieve at least a C or 73% in each course. You must monitor your grades. If you are at risk of not earning a C, you must communicate early and often with your instructor.

If you do not achieve at least a C or 73% in a class, you will earn an F. This means that you will be placed on academic probation, and that you will need to re-take the course the following year. This may delay your graduation.

Provisional Class Schedule

Lecture Date	Lecture & workshop	Assignment (focus)
#01 01/23	Descriptive statistics refresher <i>R how-to</i> : getting started and exploring data	Knowledge check, homework
#02 01/30	Inference: probabilities and uncertainty <i>R how-to</i> : complex coding with AI and practice with data wrangling	Knowledge check, homework
#03 02/06	Introduction to hypothesis testing and p-values <i>Peer review how-to</i> : introduction and focus on the statistical analyses	Knowledge check, homework
#04 02/13	Comparisons of means across two groups <i>R how-to</i> : continuous data and hypothesis testing	Knowledge check, homework
#05 02/20	Comparisons of means across several groups and analyses of variances <i>R how-to</i> : continuous data and hypothesis testing	Knowledge check, homework
#06 02/27	Comparisons of prevalences across groups <i>R how-to</i> : categorical data and hypothesis testing	Knowledge check, homework
#07 03/06	Comparing what is comparable: adjustment and standardization <i>Excel and R how-to</i> : population adjustment and standardization	Knowledge check, homework
#08 03/13	Linear and logistic regressions <i>Excel and R how-to</i> : regression models	Knowledge check, homework
#09 03/20	Studying more complex associations: multivariate models <i>R how-to</i> : multivariate models	Knowledge check, homework
#10 03/27	Beyond p-values: unmasking statistical fallacies <i>Peer review how-to</i> : looking beyond the statistics	Knowledge check, homework
04/03	Spring break	
#11 04/10	Other regression models: survival, additive, mixed... <i>Peer review how-to</i> : working around new methods	Knowledge check, homework
#12 04/17	Analyzing questionnaire data <i>R how-to</i> : working with questionnaire data	Knowledge check, homework
#13 04/24	Introduction to high-dimensional data analysis and machine learning Mini journal club	Knowledge check, homework
#14 05/01	Introduction to mechanistic modeling <i>R how-to</i> : tailoring existing models	Knowledge check, homework
- 05/08	<i>Study period</i>	Knowledge check, homework
- 05/15	<i>Exams</i>	Exam

Additional Resources

Biostatistics

- *Essentials of Biostatistics in Public Health*. Sullivan, 2023.

Basic R skills

- You need a comprehensive R reference covering R basics and complex statistical analyses: *The R Book*. Jones et al., 2022. Free PDF book via institutional login at onlinelibrary.wiley.com/doi/book/10.1002/9781119634461.
- You want a hands-on, workflow-focused guide on data analysis with R, or you want to explore modern R coding practices: *R for Data Science*. Wickham & Grolemund, 2023. Free PDF and code at r4ds.had.co.nz.
- You want to improve your data wrangling and collaborative coding skills, Cornell *NTRES 6100: Collaborative and Reproducible Data Science in R*. Overgaard Therkildsen (Cornell Natural Resources and the Environment). Free course material at nt246.github.io/NTRES-6100-data-science.
- You want to go further in epidemiological data analysis: *The Epidemiologist R Handbook*. Batra et al. 2021. Free handbook at epirhandbook.com.

Academic Integrity

Each student is expected to abide by the [Cornell University Code of Academic Integrity](#). Any work submitted for academic credit will be your own work. Any facts, ideas or data from others must be properly cited. Policies on academic integrity issues, including cheating and plagiarism, can be found in the [MPH student handbook](#). Some academic integrity violations are described below.

Your Own Work – Any work submitted by a student for academic credit must be your own work. Students are encouraged to study together and to discuss concepts and methods with other students. However, copying another student’s work or producing work for another student is considered an academic integrity violation.

Plagiarism – Copying any text, data or ideas from other sources (published, person) without appropriate citation is considered plagiarism and an academic integrity violation. Open-source code can be copied and modified, but accompanied by the appropriate citation if representing a significant proportion of the final code. Students are urged to read and complete the exercises on “Recognizing and Avoiding Plagiarism” at plagiarism.arts.cornell.edu.

Data Fabrication and Generative Artificial Intelligence (AI) – Generative AI is known to cite sources that do not exist in the real world. Submitting work with ‘fake’ data or sources (generated by generative AI or any others means) is considered an academic integrity violation.

If an academic integrity violation is identified, these are the consequences:

- On one or more of the weekly **homework** assignments:
 - *First violation*: zero on the assignment, no revision permitted. Required meeting with the instructor. Egregious first violations will be treated as a second violation.

Syllabus and course schedule are subject to change.

All changes will be indicated on Canvas.

- *Second violation*: zero on the assignment, no revisions permitted. Elevation to CVM Academic Integrity Hearing Board.
- On the **final exam**:
 - Primary academic integrity hearing with instructor and witness. If primary hearing decision is that you are responsible, zero on the assignment, no chance to re-do assignment or demonstrate mastery of competency, course failure, possible elevation to CVM Academic Integrity Hearing Board.

This course is participating in **Accepting Responsibility**, which is a pilot supplement to the Cornell Code of Academic Integrity. For details about the Accepting Responsibility process, see the [Accepting Responsibility](#) website.

Additional Notes on the Use of AI in this Course

Generative AI is a valuable tool that is increasingly used in the coding community. Therefore, appropriate and responsible use of AI for coding will be covered in this course, and students may use generative AI for assignments. However, the following applies:

- Knowledge checks:
 - The use of generative AI on the knowledge checks is strongly discouraged, as these formative assignments are designed to help you assess your knowledge.
- Homework:
 - Generative AI may be used to supplement and support your work (e.g., revising written work, grammar checking, producing base code, debugging code, etc.).
 - However (1) **AI-generated text or code should be checked**, and (2) **the work you submit must be your own**, and you should be prepared to verbally explain the work you submitted. Submitting work you cannot explain will be considered an academic integrity violation. Submitting work with fabricated evidence (generated by generative AI or any other means) is not acceptable and will be considered an academic integrity violation. You are responsible for making sure the work you turn in is accurate and correct.
 - Assignments will include an **AI use disclosure statement**. You are expected to clearly explain your use of generative AI tools and tools with AI embedded in them, such as ChatGPT, Perplexity, CoPilot, Wordtune, Quillbot, Grammarly, translators, and many others. If you are unsure of any policy or any assignment-specific directions, including whether or not a tool is considered AI, please consult the instructor prior to using the technology or completing your assignment.
 - Failure to disclose AI usage or presenting AI-generated work as solely your own without acknowledgment may be considered a breach of academic integrity. Instructors and course assistants may ask for drafts, process documentation, or oral explanation if AI use is suspected but not acknowledged.
- Final exam:
 - The use of generative AI tools will **not be permitted during the final exam**. This summative assessment is specifically designed to measure your personal mastery of data analysis and interpretation skills, with particular emphasis on your critical thinking abilities.

- In-class activities:
 - The use of **generative AI is not permitted for in-class activities** unless specified otherwise. These activities are designed to help you think critically about the content and practice applying skills, which will later be essential for you to (1) formulate effective prompts for AI, and (2) review and modify AI-generated code.

Additional Notes on the Use of AI for Coding

While AI can significantly accelerate your coding process and serve as a valuable development tool, it requires careful and responsible use. AI tools can produce code with errors, logical flaws, inefficiencies, or approaches that do not fit the specific context of your work. Additionally, AI may suggest outdated methods or fail to follow best practices for your particular use case. In professional practice, you will be held accountable for any code you submit, regardless of how it was generated, and employers expect you to understand and vouch for your work. Therefore, it is your responsibility to ensure that any AI-generated code is functional, produces the correct output, follows appropriate methodology, and that you fully understand how and why it works. This critical evaluation process is an essential skill that will serve you throughout your career and will be further covered in class.

Confidentiality and Privacy, and Accountability when Using AI

When AI is used, you are responsible for maintaining academic integrity in your use of AI tools. This includes following university guidelines related to Confidentiality and Privacy, and Accountability, as described at it.cornell.edu/ai/ai-guidelines.

- **“Confidentiality and Privacy:** If you are using public generative AI tools, you cannot enter any Cornell information, or another person's information, that is confidential, proprietary, subject to federal or state regulations, or otherwise considered sensitive or restricted. Any information you provide to public generative AI tools is considered public and may be stored and used by anyone else. As noted in the [University Privacy Statement](#), Cornell strives to honor the [Privacy Principles](#): Notice, Choice, Accountability for Onward Transfer, Security, Data Integrity and Purpose Limitation, Access, and Recourse.”
Implications: entering something into external AI tools is like posting it on a website for anyone to see.
- **“Accountability:** You are accountable for your work, regardless of the tools you use to produce it. When using generative AI tools, always verify the information for errors and biases and exercise caution to avoid copyright infringement. Generative AI excels at applying predictions and patterns to create new content, but since it cannot understand what it produces, the results are sometimes misleading, outdated, or false.”
Implications: anything wrong or misleading becomes your fault and will be treated as harshly (by the public, your colleagues, or else) as if you had fabricated yourself. See many examples in the news about this.

Professionalism

As students in a professional health degree program and community learning environment, professional interpersonal behavior is expected with your peers, community partners, and teaching team, inclusive of faculty, staff and course assistants.

One aspect of professional behavior includes participating in an active learning environment. Disruptive/unprofessional behaviors in the classroom and learning environment might include use of phones or computers if prohibited, watching online content or taking part in online activities that might distract fellow students, or using generative AI or translation software in class without explicit permission.

Respectful disagreements, differing perspectives and emotion are all a normal part of the learning environment. However, disorderly, abusive, or disrespectful behavior, whether verbal or physical in nature, will not be tolerated by anyone or to anyone.

Such behavior related to this course might result in grade penalties, course failure, or academic intervention.

- Course-related decisions (i.e., grade penalties for classroom misconduct or academic integrity violations under the Code of Academic Integrity or course failure for failure to meet learning outcomes) are up to the course faculty.
- Academic interventions and other consequences related to participation in the MPH degree training will be suggested by the faculty member and reviewed by Cornell Public Health leadership, inclusive of the Director, Associate Director, Assistant Director of Student Services, and relevant concentration area section chief.
- In addition, some offenses may also subject a student to formal Cornell non-academic disciplinary procedures under the Student Code of Conduct or Policy 6.4 and/or criminal charges.

Examples of unprofessional behavior include (but are not limited to), the following, whether on- or off-campus:

- Aggressive, threatening or intimidating speech or physical behavior, including yelling at any member of the teaching team;
- Encroaching personal space even without physical contact and failing to increase distance when asked;
- Blocking a person's movement or ability to leave;
- Refusing to leave an office or teaching team space;
- Disrupting class to the extent that teaching is interrupted;
- Comments about the capacity of any member of the learning community related to that individual's age, gender, religion, national origin/ethnicity, gender identity or sexual orientation; or
- Behavior deemed by a reasonable person as inappropriate for a professional academic setting.

Building Respect and Trust in the Classroom

Each person in this class is expected to respect the principles of academic freedom for instructors and classmates and will maintain the privacy of the classroom environment, as outlined in [Cornell's S20 Commitment to Academic Integrity, Equitable Instruction, Trust, and Respect](#).

This commitment to building respect and trust in the classroom means members of this class will not: record, photograph, or share any interactions that involve classmates or any member of the teaching team. Students will also respect the intellectual property rights of the instructor and will not share or otherwise make accessible any course materials to anyone not enrolled in the course without the instructor's written permission. In particular, **students may not use AI tools to record class or put class materials into AI tools without explicit written permission from the instructor.**

This policy is not meant to restrict students' ability to use classroom recordings in ways beneficial to their learning. Students who may benefit from recorded lectures and lecture playback, including students who use English as an additional language or who have accommodations from the Student Disability Services, should speak to the course instructor to maintain transparency and trust in the classroom. Students approved to record lectures are expected to maintain the respect and privacy of the learning environment, as stated above.

Diversity and Inclusion

Cornell supports an inclusive learning environment where diversity and individual differences are understood, respected, appreciated, and recognized as a source of strength. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. Inclusiveness is imperative to the pursuit of excellence in research, teaching, and outreach.

During this class, you may be exposed to examples of public health challenges and opportunities linked to diversity and inclusion (or lack thereof). With our collective commitment to equity, it is important that we are aware of the structural and physical barriers that result in marginalization, and that we commit to open and honest dialogue about these situations.

We must all commit to respect each other's right to opinion, to continually re-evaluate our own opinions in light of sound evidence and the perspectives of others, and to communicate our opinions in a way that shows maturity and mutual respect.

For more details on diversity and inclusion, please refer to diversity.cornell.edu

Anti-Racism & Anti-Discrimination Policy

The MPH Program is committed to pursuing an active anti-racist lens across the curriculum, engaged activities, professional development, recruitment, and other elements of our program. Any form of discrimination or harassment is strictly prohibited and will not be tolerated. Students are encouraged to inform the instructor when they experience any type of

discrimination or harassment. Students can report incidents and seek help to the [Office of Institutional Equity and Title IX](#) or [Report to Cornell](#). Every person should feel safe.

Confidentiality and Mandatory Reporting

As faculty at Cornell, one of our responsibilities is to create a safe learning environment on campus. Students should be aware that all faculty and staff at Cornell are required to report all disclosures of gender-based harassment, sexual harassment, sexual assault, domestic and dating violence, stalking, sexual exploitation, or other forms of sexual misconduct to the Title IX Coordinator of the University, as required by Cornell Policy 6.4. We will seek to keep any information you share with us private to the greatest extent possible, but would not be able to keep the information confidential given our reporting requirements to the Title IX Coordinator. If preferred, students can opt to speak to someone confidentially through any of the following resources: titleix.cornell.edu/resources/confidential.

Accommodations

Cornell University is committed to ensuring access to learning opportunities for all students. [Student Disability Services \(SDS\)](#) is the campus office that collaborates with students who have disabilities to provide and/or arrange reasonable accommodations. If you are registered with SDS and have a faculty notification letter for this semester, please contact the instructor early in the semester to review how the accommodations will be applied in the course. If you have, or think you may have, a disability, please contact the SDS office to arrange a confidential discussion regarding equitable access and reasonable accommodations. Students must register with [SDS](#) to verify their eligibility. **Requests for academic accommodation should be made during the first three weeks of the semester**, except for unusual circumstances. If you are registered with SDS and have questions or concerns about your accommodations, please contact your SDS Counselor.

Life Happens

We recognize that things come up in life. While we hope that your MPH education, and this class, are priorities, we know that many students have multiple responsibilities. We, your teaching team, are humans too. Please do not be afraid to come and talk to us if something comes up and you need some extra time or extra support. We are here.

AI Disclosure Statement

Anthropic's Claude 4 Sonnet was used to (1) review and improve class content, and (2) refine syllabus language.